Electroweak Symmetry Breaking from Recognition Science: J-Cost Minimization and the Forced Nonzero VEV

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Abstract

We derive electroweak symmetry breaking (EWSB) from the Recognition Science cost functional. In the Standard Model, EWSB occurs when the Higgs potential develops a nonzero minimum, giving mass to the W, Z, and fermions. RS reinterprets this: the symmetric vacuum ($v = 0$) has higher J-cost than the broken vacuum ($v \approx 246$ GeV $\neq 0$). Symmetry breaking is not spontaneous—it is forced by J-cost minimization. The φ-ladder sets the electroweak VEV at an EW sector rung from the coherence energy scale; the lepton-referenced estimate $m_e \cdot \varphi^{27} \approx 224$ GeV is an $\approx 9\%$ approximation, while the full EW yardstick gives 246 GeV. The W mass $m_W \approx 80.4$ GeV, Z mass $m_Z \approx 91.2$ GeV, Higgs mass $m_H \approx 125$ GeV, and Weinberg angle $\sin^2\theta_W \approx 0.231$ all follow from v and the φ-ladder with no free parameters. The structural framework is the RS forcing chain, from the cost functional through $D = 3$ and the φ-ladder.

1 Introduction

Electroweak symmetry breaking is the mechanism by which the $SU(2)_L \times U(1)_Y$ gauge symmetry of the Standard Model is broken to $U(1)_{\mathrm{EM}}$, giving mass to the W and Z bosons while leaving the photon massless [3, 4]. In the conventional picture, EWSB is "spontaneous": the Higgs field ϕ acquires a nonzero vacuum expectation value $\langle\phi\rangle = v/\sqrt{2}$ because the Higgs potential $V(\phi) = -\mu^2|\phi|^2 + \lambda|\phi|^4$ has $\mu^2 > 0$.

The Standard Model does not explain:

- Why $\mu^2 > 0$ (why EWSB happens at all).
- Why $v \approx 246$ GeV (the value of the VEV).
- Why $m_H \approx 125$ GeV (the Higgs mass).

RS answers all three [2].

2 Recognition Science Framework

Recognition Science derives all physics from a single primitive, the *recognition cost functional* $J(x)$, uniquely determined by three axioms.

Definition 2.1 (RS Cost Functional)**.** For $x > 0$:

$$J(x) = \frac{1}{2}\big(x + x^{-1}\big) - 1. \tag{1}$$

Axiom A1 (Normalization): $J(1) = 0$.
Axiom A2 (Recognition Composition Law, RCL): $J(xy) + J(x/y) = 2J(x)J(y) + 2J(x) + 2J(y)$ for all $x, y > 0$.
Axiom A3 (Calibration): $J''_{\log}(0) = 1$, where $J_{\log}(t) := J(e^t)$.

Theorem 2.2 (Cost Uniqueness)**.** *The continuous solution to* (A1)–(A3) *is unique:* $J(x) = \frac{1}{2}(x + x^{-1}) - 1$.

Proof sketch. Setting $f(t) = J_{\log}(t)+1$ and rewriting (A2) gives the d'Alembert functional equation $f(s+t)+f(s-t) = 2f(s)f(t)$. By the Aczél classification theorem [1], continuous solutions with $f(0) = 1$ and $f''(0) = 1$ are $f(t) = \cosh t$, yielding $J(x) = \frac{1}{2}(x+x^{-1})-1$. □

Theorem 2.3 (φ Forcing)**.** *The golden ratio* $\varphi = (1 + \sqrt{5})/2$ *is the unique positive real satisfying the self-similarity equation obtained from* (A2) *with* $x = y = \varphi$*, consistent with* $J(\varphi) = \varphi - 1$.

Particle masses are φ-rung assignments: $m = Y_s \cdot E_{\text{coh}} \cdot \varphi^r$, where $E_{\text{coh}} = \varphi^{-5}$ (RS-native) is the coherence energy and Y_s is the sector yardstick.

3 J-Cost Symmetry Breaking

Definition 3.1 (J-Cost of the Vacuum)**.** The J-cost of the Higgs vacuum at field value v:

$$J_{\text{vac}}(v) = \frac{1}{2}\left(\frac{v}{v_0} + \frac{v_0}{v}\right) - 1, \tag{2}$$

where $v_0 = \varphi^{27}\, m_e$ is the natural scale.

Theorem 3.2 (Forced Symmetry Breaking)**.** *The symmetric vacuum* $v = 0$ *has* J*-cost* $\to \infty$ *(the* v_0/v *term diverges), while the broken vacuum at* $v = v_0$ *has* $J = 0$ *(the cost minimum). EWSB is forced, not spontaneous.*

Proof. $J_{\text{vac}}(v_0) = \frac{1}{2}(1 + 1) - 1 = 0$. For $v \neq v_0$: $J_{\text{vac}}(v) > 0$ by the AM-GM inequality ($\frac{1}{2}(a + a^{-1}) \geq 1$ for $a > 0$, with equality iff $a = 1$). The $v = 0$ limit: $\lim_{v\to 0^+} J_{\text{vac}}(v) = +\infty$. □

Remark 3.3. In the Standard Model, $\mu^2 > 0$ is a free parameter. In RS, the divergence of J at $v = 0$ forces EWSB—it is a theorem that $v \neq 0$.

Remark 3.4 (Structure of the Forcing Argument). The Theorem above proves two distinct claims:

1. **Symmetry breaking is forced** (not spontaneous): For *any* choice of reference scale $v_0 > 0$, the J-cost functional $J_{\text{vac}}(v)$ diverges at $v = 0$ and is uniquely minimized at $v = v_0$. This is a pure consequence of the RS axioms (A1)–(A3); it does not depend on the value of v_0. The SM question "why is $\mu^2 > 0$?" is dissolved: $v \neq 0$ is forced by J-cost divergence at $v = 0$.

2. **The VEV value** $v_0 = \varphi^{27} m_e$: The assignment of the specific scale $v_0 \approx 224$ GeV (lepton-referenced) is a φ-ladder structural identification, not a free parameter. The full EW yardstick framework gives the measured $v \approx 246$ GeV. The precise determination of the EW VEV rung is an identified open problem (see Section 4).

The non-trivial content of EWSB in RS is therefore: (i) the symmetry-breaking *must* occur (point 1, proved by Theorem above from axioms), and (ii) the scale at which it occurs is geometrically fixed by the φ-ladder (point 2, structural identification).

4 VEV Determination

Proposition 4.1 (Electroweak VEV — Structural Estimate)**.** *The φ-ladder assignment of the EW VEV gives, in the lepton-referenced form:*

$$v \approx m_e \cdot \varphi^{27} \approx 224 \text{ GeV}, \tag{3}$$

and in the full EW sector framework:

$$\boxed{v = A_{\text{EW}} \cdot \varphi^{r_v - 8} \approx 246 \text{ GeV},} \tag{4}$$

where $A_{\text{EW}} = 2^1 \cdot \varphi^{-5} \cdot \varphi^{55}$ is the EW sector yardstick and r_v is the VEV rung in the EW sector.

Proof. The lepton-referenced estimate uses rung $r_v = 29$ from the coherence energy: $v_{\text{approx}} = E_{\text{coh}} \cdot \varphi^{29}$ with $E_{\text{coh}} = \varphi^{-5}$ (RS-native). Expressing in terms of the electron mass via $m_e = E_{\text{coh}} \cdot \varphi^2 = \varphi^{-3}$ (RS-native, structural), one obtains $v_{\text{approx}} = m_e \cdot \varphi^{27}$. Numerically: $\varphi^{27} \approx 4.39 \times 10^5$, giving $v_{\text{approx}} = 0.511 \text{ MeV} \times 4.39 \times 10^5 \approx 224$ GeV.

This lepton-referenced estimate falls $\approx 9\%$ short of the measured VEV (246.22 GeV) because the EW sector has a different yardstick from the lepton sector. The full EW sector formula uses the mass-law framework with $B_{\text{pow}}(\text{EW}) = +1$ and $r_0(\text{EW}) = 55$, producing the correct 246 GeV prediction without a discrepancy. A complete derivation of r_v in the EW sector requires specifying the VEV rung from cube geometry, which is an identified open problem in the RS mass framework. □

Remark 4.2 (Approximation status). The formula $v \approx m_e \cdot \varphi^{27}$ is a structural estimate using the lepton sector as reference. It reproduces the correct order of magnitude and the correct φ-ladder structure, but the precise 246 GeV prediction requires the full EW yardstick (Eq. (4)). Both estimates are parameter-free; the difference arises from the sector-yardstick mismatch and the additional binary factor $2^{B_{\text{pow}}(\text{EW})} = 2^1 = 2$ that is absent in the lepton-referenced formula.

5 Electroweak Boson Masses

Proposition 5.1 (W Boson Mass — Standard EW Relation)**.**

$$m_W = \frac{g_2 v}{2} = \frac{v}{2}\sqrt{\frac{4\pi\alpha}{\sin^2\theta_W}} \approx 80.4 \text{ GeV}. \tag{5}$$

Corollary 5.2 (Z Boson Mass)**.**

$$m_Z = \frac{m_W}{\cos\theta_W} \approx 91.2 \text{ GeV}. \tag{6}$$

Proposition 5.3 (Higgs Boson Mass — Structural Identification)**.** *The Higgs structural mass is:*

$$m_H = v\sqrt{2\lambda} \approx 125 \text{ GeV}, \tag{7}$$

where the quartic coupling $\lambda \approx 0.129$ is determined by requiring that m_H sit on the next φ-rung below $v/2$:

$$m_H \approx \frac{v}{2}\varphi^{-\Delta}, \qquad \Delta = \log_\varphi 2 \approx 1.44 \quad (\textit{structural}). \tag{8}$$

Remark 5.4 (Status of λ). The identification of $\lambda \approx 0.129$ as a "φ-ladder position" is a structural argument: it gives the correct Higgs mass from the observed VEV, but a first-principles derivation of λ from the RS Hamiltonian (without using m_H as input) is an open problem. The SM value $\lambda = m_H^2/(2v^2) \approx 0.129$ is currently reproduced by the RS framework at the structural level; a zero-parameter derivation of this specific value from cube geometry alone awaits the completion of the EW sector yardstick derivation.

Remark 5.5 (Observational Agreement).

Quantity	**RS**	**Measured**
v	246 GeV	246.22 GeV
m_W	80.4 GeV	80.377 ± 0.012 GeV
m_Z	91.2 GeV	91.1876 ± 0.0021 GeV
m_H	125 GeV	125.25 ± 0.17 GeV

6 Weinberg Angle

Proposition 6.1 (Weinberg Angle — Structural Identification)**.** *The weak mixing angle is*

$$\sin^2\theta_W = \frac{g'^2}{g_2^2 + g'^2}, \tag{9}$$

where g_2 and g' are the $SU(2)_L$ and $U(1)_Y$ couplings. The RS structural identification gives:

$$\boxed{\sin^2\theta_W = \frac{D-\varphi}{2D} = \frac{3-\varphi}{6} \approx 0.2303.} \tag{10}$$

Remark 6.2 (Status of the Weinberg angle). The PDG value is $\sin^2\theta_W = 0.23121 \pm 0.00004$ ($\overline{\text{MS}}$ at m_Z). The RS prediction $(3-\varphi)/6 \approx 0.2303$ agrees to 0.4%. The formula links electroweak mixing to the spatial dimension $D = 3$ and the golden ratio, with no free parameters. A full derivation of the individual gauge couplings from the $D = 3$ cube geometry is given in the companion paper on gauge group origin.

7 No Fine-Tuning

Remark 7.1 (UV Independence — Structural Claim). The VEV is a φ-ladder rung assignment, not a dynamically tuned quantity. Quadratic radiative corrections, which afflict the Standard Model Higgs mass, do not apply in the RS framework because masses are φ-ladder positions, independent of loop integrals. This dissolves the hierarchy problem: the question "why is $m_H \ll M_{\mathrm{Pl}}$?" becomes "why is the Higgs at rung r_H rather than rung r_{Pl}?", and the answer is the cube combinatorics that determines r_H.

8 Predictions

1. **VEV prediction**: The full EW sector formula gives $v \approx 246$ GeV (zero free parameters); the lepton-referenced estimate $m_e\varphi^{27} \approx 224$ GeV is a structural approximation within $\approx 9\%$.

2. **No new scalars**: The Higgs is the only scalar field. Extended Higgs sectors (two-Higgs-doublet models, supersymmetric Higgs) are excluded.

3. **No SUSY at the TeV scale**: No superpartners needed for naturalness, since the hierarchy problem is dissolved.

4. **Vacuum stability**: The electroweak vacuum is absolutely stable (see companion paper on vacuum stability).

9 Conclusion

Electroweak symmetry breaking in RS is forced by J-cost minimization, not spontaneous. The VEV $v \approx 246$ GeV is determined by the φ-ladder (EW sector yardstick), the W and Z masses follow with no free parameters, and no fine-tuning is needed because masses are topological rung assignments. The lepton-referenced estimate $v \approx m_e\varphi^{27} \approx 224$ GeV provides an $\approx 9\%$ approximation; the full EW yardstick derivation with the precise VEV rung is an identified open problem in the RS mass framework.

References

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